24/7 ZERO-LATENCY HEALTHCARE AI

CareVoice AI



Autonomous Dental Clinic Voice Receptionist & Live Practice CRM

Eliminating missed appointments with ultra-low latency voice streaming, Google Calendar, Sheets & WhatsApp automation

ABSOLUTE DENTAL CLINIC, LAS VEGAS, NV · ENGINEERING TEAM: CAREVOICE AI

Made with GAMMA

THE PROBLEM

Dental practices lose 30%+ of high-value appointments

30%+

Missed Calls

Patients call after 5 PM and on weekends when the clinic is closed, and book with competing practices.

$150,000+

Annual Leak

Unfilled chairs, last-minute cancellations, and scheduling friction hurt profitability.

Overload

Staff Overload

Receptionists juggle triage, check-ins, and ringing phones, leading to burnout and error-prone bookings.

Made with GAMMA

MEET SOPHIA

Sophia is the 24/7 autonomous voice receptionist



< 300ms

Turnaround: instant voice response that feels human.

24/7/365

Available: never sleeps, takes lunch breaks, or misses a ring.

Natural Spoken Times

Says slots as "2 PM" or "11 AM" with no robotic stutter.

Live Slot Triage

Verifies actual doctor availability before offering openings.

Full Lifecycle

Books, reschedules, and cancels appointments on its own.

Multilingual

Fluent in English and Spanish.

Made with GAMMA

ARCHITECTURE

A production-grade real-time pipeline

01

Speech-to-Text

AssemblyAI Universal-3.5 real-time streaming WebSocket, 16kHz PCM audio with instant turn detection.

02

Reasoning Brain

OpenAI GPT-4o-mini for clinical intent parsing, dental guardrails, and deterministic tool calls.

03

Speech Synthesis

Cartesia Sonic-2 TTS, 24kHz studio audio, sub-150ms voice generation, instant barge-in interruption.

04

Clinic Automation

n8n webhook engine on Railway, two-way sync with Google Calendar, Sheets, and WhatsApp.

Made with GAMMA

AUTOMATION

Automation goes well beyond voice

Smart Rescheduling & Cancellation

Finds existing appointments by patient name and phone number, releases the old slot, and locks in the new time.

Automated WhatsApp Confirmations

Instant personalized message on booking or rescheduling with patient name, date, time, and Absolute Dental clinic address.

Two-Way Calendar & EHR Sync

Google Calendar and Sheets update in real time, eliminating manual data entry for staff.

Made with GAMMA

DASHBOARD

Real-time operational visibility for clinic staff



Live Telemetry Monitor

Green health indicators for AssemblyAI STT, OpenAI LLM, Cartesia TTS, and the n8n engine.

AI Call Transcripts & Auditing

Historical logs of call duration, patient intent, execution status, and verbatim transcripts.

Visual Monthly Practice Calendar

Interactive grid with patient chips, working hours, and status filters.

Role-Based Access Security

Protected with clinic biometric/PIN security (2026) and admin session management.

Made with GAMMA

RESULTS

Quantifiable clinic results and value created

100%

Inbound call capture, with zero missed opportunities day or night.

< 300ms

Conversational response, indistinguishable from a human.

30%+

Recaptured practice revenue from previously lost after-hours inquiries.

99.9%

Autonomous scheduling accuracy through direct calendar integration.

Made with GAMMA

LIVE NOW

Deployed and production-ready

LIVE PATIENT VOICE PORTAL

https://dental-care-ai-voice-agent-production.up.railway.app/

CLINIC OPERATIONS CRM DASHBOARD

https://dental-care-ai-voice-agent-production.up.railway.app/dashboard/

GITHUB SOURCE REPOSITORY

https://github.com/Mahinn849/dental-care-ai-voice-agent

DEPLOYMENT INFRASTRUCTURE

Fully containerized with Docker, running on Railway Cloud.

Made with GAMMA